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Fluids and Barriers of the CNS

REVIEW

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Invertebrate glial barriers as a model for understanding blood–brain barrier evolution

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Abstract

Biological barriers play a crucial role in maintaining tissue homeostasis across diverse animal taxa, from invertebrates to mammals. In the nervous system, they regulate ion balance, metabolic exchange, and immune protection, ensuring proper neuronal function. In arthropods, the blood–brain barrier (BBB) is primarily formed by the perineurium, consisting of perineurial and subperineurial glial cells that establish septate junctions to restrict diffusion. Cephalopods, such as octopuses and squids, possess two distinct BBBs: one formed by glial cells and another by pericytes, depending on the type of brain blood vessel. Similarly, in vertebrates such as sharks, skate, rays, and sturgeons, the BB is also formed by glial cells. In contrast, the BBBs of most vertebrates rely on endothelial tight junctions, although astrocytes and pericytes contribute significantly to BBB maintenance and function. Importantly, glial barriers also exist in vertebrates, including the blood-nerve barrier (BNB), and the blood-cerebrospinal fluid barrier (BCSFB). Despite structural differences, the molecular mechanisms governing barrier formation, function, and plasticity show remarkable evolutionary conservation between invertebrates and vertebrates. In this review, we examine the diversity of glial barriers, their structural and functional parallels, evolutionary origins, and the key molecular pathways that regulate their development.

Keywords *Drosophila melanogaster*, Blood–brain barrier, Glial cells, Glial barrier

Background

The establishment of biological barriers that separate compartments is fundamental to the organisation of complex organisms. Even the simplest multicellular animals rely on epithelial barriers to distinguish their internal environment from the external surroundings. In more advanced organisms, organs and tissues are often compartmentalised and isolated to maintain specialised functions. These barriers are critical for regulating physiological processes and ensuring proper organisation within the body.

Biological barriers not only serve as physical and chemical shields but also serve a critical role in facilitating and regulating communication between cells or compartments in an organism. These dynamic interfaces selectively control the exchange of ions, nutrients, metabolites, and signalling molecules while protecting sensitive tissues from toxins, pathogens, and external fluctuations [1, 2]. Maintaining a balance between isolation and communication is vital for physiological processes and coordination across organ systems. To achieve this, the cellular components of biological barriers must be tightly regulated, enabling a collective and efficient response to environmental changes or special internal organ demands.

In the nervous system, several barriers protect neural tissue from external influence. Among these, the blood–brain barrier (BBB) and the blood-cerebrospinal fluid barrier (BCSFB) constitute key borders between the neural tissue and the circulatory system [3]. More than a century ago, Paul Ehrlich, and later Edwin Goldmann, injected dyes into the circulatory system, finding that

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they permeated most organs but were excluded from the brain parenchyma [4, 5]. However, the concept of a physical interface or barrier was not formalised until the early twentieth century [6–11]. Both BBB and BCSFB control the transport of essential molecules, such as glucose and amino acids, into the brain while restricting the entry of harmful substances [12, 13].

The metabolic support within the nervous system is generally considered to be mediated by a specialised group of cells, collectively known as glial cells. Originally described as "glue cells" due to their presumed structural role of "*filling space*", glial cells (or simply glia) are recognised as key players in a wide range of functions [14]. Beyond providing structural support, glial cells are critical for preserving brain homeostasis, regulating neurotransmitters, facilitating waste removal, orchestrating immune responses, and supporting various other processes [15]. In many cases, glial cells are integral to the formation and maintenance of barriers in the nervous system. A prominent example of this glial function is the involvement of astrocyte end-feet in regulating BBB permeability, as well as their contribution to the function of the glymphatic system [16].

Across different taxa, glial cells serve as the principal cell type responsible for forming tight barriers. While the evolutionary pressures driving glial cells to fulfil this crucial role remain barely understood, it is evident that this strategy has been repeatedly and successfully employed multiple times throughout the evolution of species [17]. In this review, we compile both classical and recent evidence showing that glial cells behave as architects of tight barriers in the nervous system across diverse animal groups. Specifically, we focus on analysing the shared characteristics of glial barriers in invertebrates such as arthropods and cephalopods, and we compare them to those found in elasmobranchs, sturgeons, and mammals (Table 1).

Main text

The blood–brain barrier of invertebrate animals

On Earth, invertebrates represent the most numerous and diverse group of animals [18]. With remarkable anatomical variety across numerous phyla, nervous systems are present in organisms ranging from Ctenophora (Comb jellies) to more complex invertebrates, though these systems have arisen through multiple evolutionary events [19, 20]. The increase in complexity of the nervous system correlates with enhanced information-processing functions and cognitive capabilities [21, 22]. This complexity likely exerted a strong evolutionary pressure for the emergence of barriers that isolate the brain parenchyma from the circulatory system, thereby protecting it from fluctuations in ion, metabolite, and nutrient concentrations [21–25]. Among invertebrates, two major groups -arthropods and cephalopods- independently evolved occluding brain barriers, which may highlight the role of convergent evolution in maintaining neural homeostasis.

The perineurium in the arthropod glial barrier

During the twentieth century, the existence of a BBB in arthropods was first proposed in studies on potassium diffusion in locusts [26]. Insects have a high but variable potassium concentration in their haemolymph, which could disrupt neuronal function. This led to the hypothesis that their nervous system required a mechanism for the regulation of ion homeostasis, independent from the haemolymph. Early studies described the insect BBB as a thin connective tissue layer that slowed potassium diffusion [26–30]. This structure, later referred to as the "*perineurium*", was subsequently identified as a layer of glial cells [31, 32]. Covering these cells is an acellular, collagen-rich extracellular matrix known as the "neural lamella" [23, 33].

Later, the BBB was further characterised using transmission electron microscopy in multiple insect species [27]. Its functional role in blocking the diffusion of high-molecular-weight tracers was experimentally demonstrated in cockroaches, moths, and stick insects [32, 34–37]. This sealing capability of the perineurium was attributed to specialised cell adhesions, originally described as "septate desmosomes" and now recognised as pleated septate junctions [38–42]. Unlike vertebrate tight junctions, which are formed by close membrane contacts known as "kissing-points", septate junctions are a type of occluding junction found across invertebrate metazoans. They exhibit a leader-like, electron-dense septa structure that forms a circumferential belt around epithelial cells. Septate junctions are located basally to adherens junctions, whereas tight junctions occupy an apical position [43, 44]. Importantly, although both types of junctions serve a similar barrier function, their molecular components evolved independently [40].

Subsequent studies demonstrated the presence and function of occluding barrier in various other arthropod taxa, including arachnids such as scorpions and spiders [45, 46], myriapods such as centipedes and millipedes [47, 48], and crustaceans such as crabs and crayfish [21, 49–53]. However, some chelicerates, like ticks and horseshoe crabs (Limulus), lack a fully sealed BBB [21, 54–56]. Consequently, it has been proposed that within the phylum Arthropoda, the BBB evolved independently in two distinct lineages: crabs/insects and spiders/scorpions [22].

Table 1 Cell types forming brain barriers in different taxa

Phylum Class	Species	Cell type	Tight junction	Septate junctions	Gap junctions	General functions
Arthropoda, Insecta	*Drosophila melanogaster*	Subperineurial glia (SPG)	✗	✓	✓	Transcellular transport, block paracellular diffusion, xenobiotic efflux
		Perineurial glia (PG)	✗	✗	✓	Carbohydrate metabolism, circadian rhythm, sleep regulation
		Ensheathing glia	✗	✗	✓	Neuropil non-occluding barrier
		Tract glia	✗	✗		Form the lacunar system (non-occluding)
		Wrapping glia	✗	✗	✓	Axon ensheathing (non-occluding)
Mollusca, Cephalopoda	*Sepia officinalis, Octopus vulgaris*	Endothelial cells (EC)	✗	✗		Fenestrated blood vessels
		Pericytes	**UD**	**UD**		Form occluding junctions in arteries and arterioles
Chordata, Actinopterygii Chordata, Dipnoi	*Huso gueldenstaedtii, Protopterus annectens*	Perivascular glia	**UD**	**UD**	✓	Form occluding junctions in veins, venules, and capillaries
		endothelial cells	✗	✗		Fenestrated blood vessels
		Pericytes	✗	✗		
		Astrocytes	✓	✗	✓	Form occluding junctions in all brain blood vessels
Chordata, Mammalia	*Mus musculus*	Endothelial cells (EC)	✓	✗	✓	Form occluding junctions in most brain blood vessels (except in circumventricular organs).
		Smooth muscle cells	✗	✗	✓	Vasoconstriction
		Pericytes	✗	✗	✓	Paracellular permeability regulation
		Astrocytes end-feet	✗	✗	✓	Metabolic regulation, synaptic modulation
		microglia	✗	✗		Immune
		α-Tanycytes	✗	✗	✓	CSF-parenchyma barrier (non-occluding)
		β1-Tanycytes	✓	✗	✓	ARC parenchymal barrier
		β2-Tanycytes	✓	✗	✓	BCSFB, transcellular transport of nutrients at the ME
		Oligodendrocytes	✗	✗	✓	CNS axon insulation
		Schwann´s cells	✗	✓		Peripheral nerve myelin, axon insulation
		Perineurial cells	✓	✗		Nerve-blood barrier
		Endoneurial EC	✓	✗		Nerve-blood barrier
		Epineurial EC	✗	✗		Fenestrated blood vessels

UD: undetermined, ✗: not present, ✓: present

The fruit fly as a model for understanding the development and function of the glial barrier

The study of model organisms has contributed enormously to our understanding of the nervous system development, physiological function, and aging. Among invertebrates, the fruit fly *Drosophila melanogaster* and the nematode *Caenorhabditis elegans* are the most extensively studied models [57–59]. Due to the powerful genetic tools available for analysing gene function and for labelling specific cell types, most of our current knowledge about the arthropod BBB has been derived from research in *Drosophila*. However, this reliance on a single model organism may introduce biases, as certain characteristics observed in *Drosophila* may not be representative of all arthropods.

In *Drosophila*, as in other insects, the BBB is characterised by a perineurium that restricts the diffusion of high-molecular-weight tracers [60–62]. The insect BBB is formed by two different layers of glial cells, collectively referred to as surface glia, which encase the entire nervous system [63]. The inner layer, composed of subperineurial glial cells (SPG), establishes the paracellular barrier through the formation of pleated septate junctions. These subperineurial glial cells are large, flat cells that share functional similarities with vertebrate endothelial cells. Overlying subperineurial glia, a second monolayer of perineurial glial cells (PG) directly contacts the neural lamella and the haemolymph, providing additional protection and regulatory functions (Fig. 1) [24, 63–66].

The formation of the *Drosophila* glial barrier begins during embryogenesis, establishing a sealed structure before larval hatching. Embryonic neuroblasts (the *Drosophila* neural stem cells) generate a fixed number of subperineurial glial cells [72, 73], which migrate over the developing nervous system and, through a mesenchymal-epithelial transition, form a squamous secondary epithelium. During late embryonic development, these cells establish septate junctions to create the paracellular barrier [74]. Mutations in core septate junction components result in leaky barrier phenotypes, highlighting their key relevance in preventing paracellular diffusion [66, 75–79].

The developmental origin of perineurial glia was initially misattributed to the mesodermal lineage [80, 81]. While their precise lineage remains unclear, only a few perineurial glial cells are generated during embryogenesis, and they do not fully enclose the nervous system, leaving the subperineurial layer in direct contact with the neural lamella [63, 74]. This interaction between subperineurial glia and the extracellular matrix is thought to provide the initial cues for subperineurial glial polarisation [74]. During larval development, perineurial glial cells

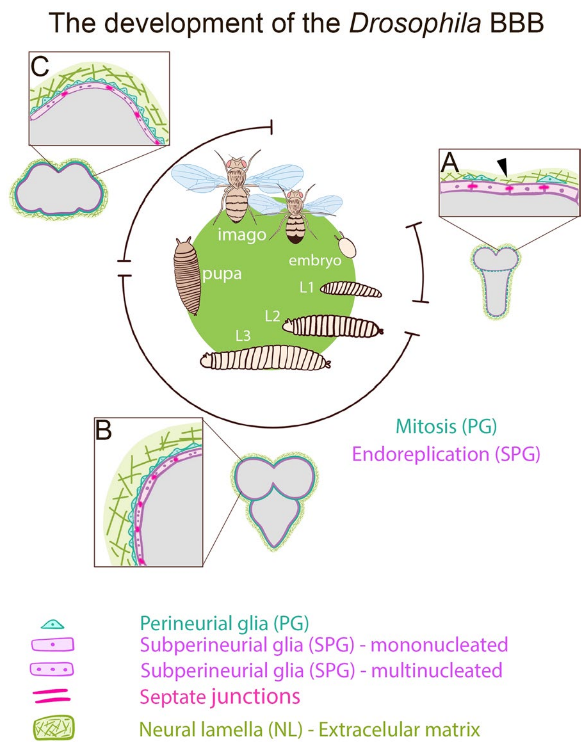

Fig. 1 The development of the *Drosophila* blood–brain barrier. A schematic representation of the *Drosophila* life cycle and the formation of the blood–brain barrier (BBB) during development. BBB of the CNS of **A** late embryo/first instar larva, **B** third instar larva, and **(C)** adult fly. Subperineurial glia (SPG) establish septate junctions at the end of embryogenesis, forming the BBB. During larval development, perineurial glia (PG) proliferate to form the second glial layer, separating the SPG from the neural lamella, an extracellular matrix. Meanwhile, SPG increase in size through endoreplication, endocycle, and endomitosis, until metamorphosis. The most striking differences between early larval and adult BBB include: the fat body secretes an unknown factor into the haemolymph that activates SPG secretion of insulin-like peptides to reinitiate neurogenesis during first instar larval stage, the number of SPG nuclei dramatically increases during late larval development, PG amplify their number during larval development, and PG morphology changes from a star-like shape in the larval BBB to an elongated form in the adult brain [63, 66–71]. Arrowhead points the contact between the subperineurial layer to the neural lamella during early larval development. Grey zone corresponds the *Drosophila* CNS cortex and neuropils

undergo mitosis, increasing in number to form a distinct layer between the neural lamella and the subperineurial glial cells [66, 67] (Fig. 1).

The genetic mechanisms governing *Drosophila* BBB formation during embryogenesis are regulated by G-protein coupled receptor (GPCR) signalling, which plays a crucial role in coordinating cellular interactions and ensuring proper septate junction formation. Loss of the orphan GPCR *moody* disrupts BBB paracellular permeability by altering cell–cell adhesion [82, 83]. In *moody* mutants, septate junctions appear frayed and fragmented, while a compensatory mechanism increases the cell–cell contact areas and potentially decreases the leakiness of the barrier [82, 84–86]. This compensatory effect may also underlie the subperineurial glial membrane overgrowth observed in *moody* mutants [87].

The establishment of subperineurial glia polarity is essential for the BBB structural integrity. During development, Moody localises at the plasma membrane facing the nervous system (sometimes referred to as the neural or apical side), where it activates the Gαi, inhibiting Adenylyl cyclase and reducing 3',5'-cyclic AMP (cAMP) levels [82]. In contrast, on the plasma membrane facing perineurial glia (humoral or basal side), GPCR/Moody signalling is reduced, resulting in elevated cAMP levels and increased Protein kinase A (PKA) activity, which regulates actin dynamics [84, 86, 88]. Both loss and gain of GPCR/Moody signalling disrupt glial polarity, mislocalising PKA and impairing septate junction formation [84]. Similarly, *moody* mutants exhibit mislocalisation of ATP-binding cassette (ABC) transporters, such as Multi drug resistant 65 (Mdr65), the *Drosophila* homologue of P-Glycoprotein [87]. These findings suggest that proper subperineurial glial polarity is essential for regulating both paracellular and transcellular transport, ensuring BBB structural integrity and function.

After larval hatching, the BBB is fully formed, but a second round of neurogenesis significantly increases the size of the nervous system. It is proposed that this growth drives subperineurial glial cell size expansion. Instead of undergoing mitosis, subperineurial glial cells increase in size through endoreplication (endocycle and endomitosis) [66, 67, 89–93]. In coordination with cellular growth, septate junctions are able to stretch with minimal protein turnover, maintaining the structural integrity of the BBB [86]. However, uncontrolled growth, such as that induced by brain tumours, can compromise barrier function [86, 90].

During metamorphosis, the BBB preserves its cellular composition and structural integrity into adulthood. At the onset of pupal development, activation of the immune deficiency (IMD) pathway can trigger immune cell infiltration into the nervous system [94]. The IMD pathway induces the secretion of interleukin-like unpaired ligands, which activate the JAK/STAT signalling and trigger neuroinflammation. Among its downstream targets, matrix metalloproteases Mmp1 and Mmp2 are upregulated, promoting neural lamella degradation and enabling macrophage invasion into the CNS via a yet unclear mechanism [95]. Notably, subperineurial glial paracellular permeability is also compromised under pathological conditions, including traumatic brain injury, bacterial infections, and metastatic tumours [96–99]. Inflammation is a common factor among all these conditions, suggesting a potential link between immune signalling and BBB structural integrity. Supporting this idea, mutant animals with a compromised BBB, like *moody* mutants, exhibit an inflammatory state [85], reinforcing the hypothesis that BBB breakdown and inflammation may exacerbate each other in a detrimental feedback loop.

The *Drosophila* glial barrier as a metabolic centre

The concept of biological barriers as passive structures that merely restrict molecular transport is an oversimplification. The *Drosophila* glial barrier not only serves as a protective structure but also functions as a dynamic metabolic centre. Through specialised transport mechanisms, metabolic enzymes, and communication with other glial subtypes, the *Drosophila* glial barrier orchestrates systemic metabolic homeostasis, adapting neural activity to physiological demands [25, 64, 100].

Since subperineurial glia act as the primary barrier, essential nutrients such as amino acids, lipids, carbohydrates, and other metabolites must traverse this layer to reach the nervous system. Consequently, both paracellular and transcellular transport of nutrients are tightly regulated at the level of subperineurial glial cells. However, transport and other physiological functions of the *Drosophila* BBB are also influenced by perineurial glial cells. One noteworthy example is the expression of the key trehalose transporter Tret1-1, which is specifically localised in perineurial glial cells [101]. Trehalose, the predominant disaccharide in the insect haemolymph, is present at significantly higher concentrations than glucose. Thus, it is likely that systemic trehalose is metabolised into glucose or glycolytic intermediates within perineurial glial cells before being transported into the nervous system. Other carbohydrate transporters, such as Pippin and MFS3, are also expressed in the BBB [102]. In addition, the BBB contains amino acid transporters [103, 104] and lipoprotein receptors [105, 106], further underscoring its role in nutrient regulation.

During larval development, neurogenesis resumes only when larvae ingest essential amino acids [68]. Subperineurial glia act as nutrient sensors in this process, releasing insulin-like peptides (dIlps) in response to an amino acid-rich diet. These dIlps subsequently activate the growth and proliferation of larval neuroblasts, initiating their reactivation [69, 70]. This response is coordinated through communication within the subperineurial glial network. Notably, subperineurial glial cells are interconnected by gap junctions composed of innexin proteins [107, 108]. Dietary amino acids trigger the release of an unknown fat body-derived signal, which promotes cytoplasmic calcium release in subperineurial glia, ultimately coordinating dIlp secretion into the brain cortex [107, 109]. Additionally, TGF-β signalling, mediated by the BMP ligand Glass bottom boat (Gbb), also regulates communication between neuroblast and perineurial glia, promoting neurogenesis [110].

Subperineurial glia are also sensitive to nutrient restriction during development. The expression of the serine protease homologue *scarface* (*scaf*) in subperineurial glia is regulated by larval nutritional status, controlling both BBB growth and neurogenesis [111]. TGF-β signalling is increased under larval starvation, upregulating the expression of the Tret1-1 in perineurial glial cells of the BBB and promoting the carbohydrate influx into the CNS [112]. In addition to carbohydrate transport, subperineurial glia also display high levels of monocarboxylate transport, measured by lactate influx, which may be necessary for maintaining neuronal activity [113]. Recently, it has been shown that proton-coupled monocarboxylate transport is increased in larval subperineurial glia under starvation conditions [114]. This starvation response is mediated by the upregulation of a novel monocarboxylate transporter CG8041/Yarqay in subperineurial glia [114]. This role as a metabolic centre is essential for maintaining brain development in *Drosophila*, in a similar way to the "brain sparing" phenomenon observed in human intrauterine growth restriction [115–119]. All this evidence highlights the BBB's ability to dynamically adjust to nutrient scarcity, regulating neurogenesis and ensuring an adaptive response to metabolic stress.

Cell–cell communication at the blood–brain barrier modulates complex behaviours in *Drosophila*

Cell-to-cell communication ensures that tissues function as a cohesive unit, synchronising their activities to maintain homeostasis, respond to stimuli, and execute complex physiological functions. The *Drosophila* BBB can operate as an integrated structure, coordinating dIlp release during development [109]. However, this collective behaviour extends beyond development and plays a crucial role in broader physiological processes in the adult fly.

Circadian rhythms are physiological behaviours that occur in daily cycles and are regulated by internal and external cues. The circadian rhythms are driven by highly conserved mechanisms known as the circadian clocks or pacemakers [120, 121]. The molecular components of the circadian clock, composed of transcriptional feedback loops, are also present in perineurial glia. In adult flies, the circadian clock regulates BBB permeability to xenobiotic agents, increasing at the onset of the night [122, 123]. The rhythmic regulation of BBB permeability does not rely on gene expression but rather on the activity of ABC transporters in subperineurial glia.

Communication between perineurial glia, where the circadian clock resides, and subperineurial glia, where the xenobiotic Mdr65 transporter is expressed, is mediated by Innexin-1 and -2 gap junctions [122]. The expression of innexin genes also follows a daily cycle, resulting in oscillations in ionic coupling between perineurial and subperineurial glia [122]. During nighttime, Mg^{2+} concentrations decrease in subperineurial glia, reducing the activity of xenobiotic transporters and, as a consequence, increasing permeability to organic compounds. Conversely, during the daytime, gap junction-mediated coupling between BBB glial cells is reduced, leading to an increase in Mg^{2+} levels in subperineurial glial cells and elevating the efflux of organic molecules out of the fly nervous system [122].

Sleep is an essential physiological state of animals, and in adult *Drosophila,* it is regulated by multiple factors, including the circadian pacemakers [120, 124]. The function of the BBB has been extensively associated with the role of sleep. Interestingly, *mdr65* null mutant flies exhibit longer sleep duration compared wild-type counterparts, a phenomenon attributed to the impaired efflux of the steroid hormone ecdysone (20-hydroxyecdysone)—known to promote sleep—out of the brain [125–127]. Although steroid hormones can cross the plasma membrane, ecdysone influx across the BBB requires the action of the Ecdysone Importer (EcI/Oatp74D) [128–130]. This suggests that the balance between clearance and molecule efflux out of the BBB is essential for sleep–wake homeostasis.

Similar to *mdr65* loss, impairment of endocytosis in subperineurial glia increases total sleep duration in *Drosophila* [131]. During sleep, endocytosis and vesicle trafficking are enhanced, facilitating the elimination of acylcarnitine fatty acids from the brain [131, 132]. This process requires the function of lipoprotein receptors LRP1 and LRP2/Mgl, as well as organic cation transporters in the BBB glial cells. Knockdown of these transporters leads to increased sleep [132], supporting the notion that sleep promotes the clearance of brain metabolites via the transcytosis route. Additionally, the carnitine transporter CG6126, expressed in subperineurial glia, mobilises peripheral carnitine into the brain. Reduction in the expression of this transporter increases wakefulness, suggesting that fatty acid metabolism drives sleep and that the net BBB permeability to carnitine regulates the sleep–wake balance [133]. Together, this evidence highlights the physiological significance of BBB transcytosis in *Drosophila*. However, it remains unclear whether transcytosis is developmentally regulated in the *Drosophila* BBB, as it is in zebrafish and mice, where suppression of this process is critical for maintaining low BBB permeability [134–137].

In addition to transcellular transport, paracellular permeability also contributes to the sleep–wake balance. GPCR signalling mutants, which display impaired septate junctions, exhibit increased sleep. Interestingly, BBB paracellular permeability and sleep are reciprocally regulated. Sleep deprivation has been shown to increase the paracellular permeability of high-molecular-weight tracers across the barrier [138]. However, the specific molecules involved in sleep regulation through this pathway remain unknown, and it is plausible that the anomalous diffusion of multiple molecules affects sleep homeostasis.

Beyond sleep regulation, the function of BBB glia influences neuronal activity. For instance, seizures, which are characterised by uncontrolled neuronal electrical activity, can be induced in *Drosophila* by mechanical or thermal stress [139]. BBB dysfunction has been extensively associated with epilepsy and seizures [140]. Interestingly, perineurial glia display calcium waves in the adult central nervous system, which are propagated by gap junctions. These waves depend on calcium storage within the endoplasmic reticulum and are essential for preventing heat shock-induced seizures in adult flies [141]. The precise mechanism by which calcium waves in the BBB influence neuronal excitability remains unclear. However, given that a primary role of the BBB is to regulate ion homeostasis and maintain the proper microenvironment for neuronal activity [23], it is unsurprising that disruptions in glial barrier communication can lead to altered synaptic transmission.

Cell–cell communication within the adult *Drosophila* blood–brain barrier regulates key physiological processes such as sleep, circadian rhythms, and neuronal activity by dynamically controlling permeability and metabolic balance. These findings suggest that glial barriers actively shape brain function, highlighting potential evolutionary parallels with mammalian systems.

The glial barrier in Cephalopods

In the phylum Mollusca, the BBB appears to have independently evolved in a single class: Cephalopoda. Octopuses and squids are remarkable animals, renowned for their complex behaviours and vertebrate-like cognitive capabilities. Unlike other classes like Gastropoda, which lack an occluding barrier between their neural tissue and circulatory system [17, 22, 23]. Cephalopods exhibit this unique feature. Furthermore, cephalopods possess a closed vascular system, similar to that of annelids and chordates, as well as a multichambered heart [142].

The presence of the BBB in Cephalopods has been functionally demonstrated through the injection of radiolabelled tracers in the cuttlefish, *Sepia officinalis* [143, 144]. In both cuttlefish and octopus brains, blood capillaries are composed of discontinuous endothelial cells and pericytes [145, 146], which are surrounded by their respective collagen basement membranes and entirely ensheathed by perivascular glia [147–150]. At the subcellular level, the sepia BBB is also established by restricting junctions; however, their ultrastructure does not resemble either vertebrate tight junctions or arthropod septate junctions [147, 149, 151, 152].

Using horseradish peroxidase (HRP) injection combined with transmission electron microscopy, researchers observed peroxidase activity at the level of pericytes in large arterioles and arteries. In contrast, in veins, HRP remained permeable to pericytes, and the BBB is established by the perivascular glia. These findings imply that in *Sepia*, the BBB is formed by different cell types depending on the vascular type [148] (Fig. 2). However, the classification of cephalopod "pericytes" could be debated. Vertebrate pericytes are an heterogeneous population that develops from neural crest cells and hematopoietic tissue [153–155]. Given that cephalopods lack neural crest-derived lineages, it is unclear whether these perivascular cells can be truly equated to vertebrate pericytes. This distinction is crucial for understanding whether cephalopod vascular structures

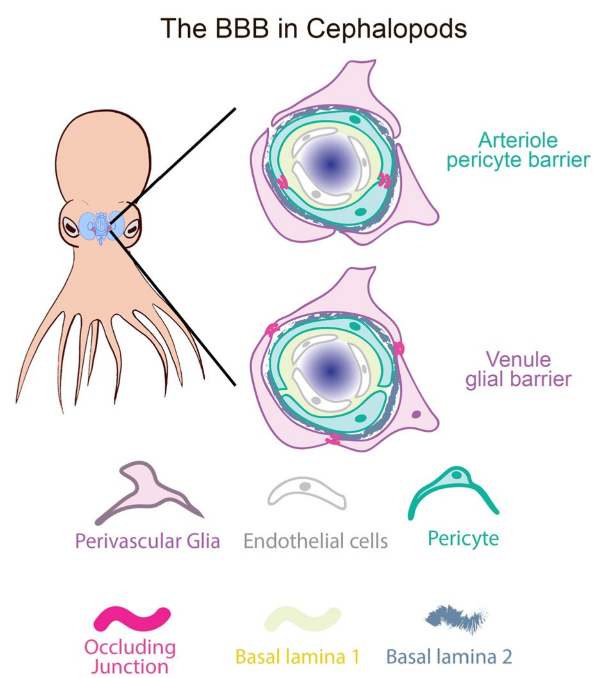

Fig. 2 The blood–brain barrier of cephalopods. A diagram illustrating the two types of barriers that separate the circulatory system from the nervous tissue in cephalopods. In arteries and arterioles, pericytes form the primary blood–brain barrier (BBB) by the formation of occludins junctions that prevent paracellular permeability. In contrast, in veins, venules, and capillaries, occluding junctions are established at the glial level, surrounding both pericytes and endothelial cells

are homologous to those in vertebrates or may represent a glial subtype.

Given the evolutionary distance between cephalopods and vertebrates, an important question arises: Does the glial barrier in cephalopods more closely resemble the vertebrate endothelial barrier or the arthropod subperineurial barrier? Permeability analyses indicated that *Sepia* microvessels exhibit permeability to non-electrolytes like that of rat endothelial cells [21]. Using a different approach, Zhang et al. analysed the *Octopus vulgaris* CNS transcriptome and proposed that the expression of Claudins, JAM, and ZO-related proteins supports a resemblance to the vertebrate BBB [156].

Recently, single-cell transcriptomic analysis of the developing *Octopus vulgaris* brain has identified three subtypes of glial cells [157]. Notably, all these glial clusters express the ETS family transcription factor Pointed, highlighting a similarity with *Drosophila* glia [158]. The cluster Glia subtype 1 (GLIA1) exhibits gene expression patterns similar to *Drosophila* ensheathing glia and mouse astrocytes, classifying it as neuropil glia that may give neuron structure and metabolic support. These cells may play a role in axon wrapping and synapse infiltration. Glia subtypes 2 and 3 (clusters GLIA2/3) are localized in the plexiform layers of the optic lobe and in a layer surrounding the brain, respectively, suggesting that the latter may have a role as part of the BBB [157, 159]. Other single-cell RNA sequencing analyses in *Octopus bimaculoides* and bobtail squid *Euprymna berryi* have identified glial clusters, but their barrier properties remain undescribed [160, 161]. Interestingly, no pericyte cluster was described from single-cell transcriptomic analysis [157, 160, 161].

Cephalopod perivascular glia may share more structural similarities to vertebrate astrocytes than *Drosophila* subperineurial glia. Although transcriptomic studies provide valuable insights, further research is needed to clarify whether cephalopod glial barriers operate more like vertebrate astrocytes or arthropod perineurial glia. Comparative genomic and ultrastructural analyses will be essential in unravelling the evolution of blood–brain barrier mechanisms across diverse animal lineages.

The vertebrate brain barriers

Given the anatomical and functional complexity of the vertebrate nervous system, it is not surprising that multiple interfaces exist between neural tissue and the circulatory system. The BBB, the primary exchange site of macromolecules entering and leaving the brain, is part of a larger functional structure known as the neurovascular unit (NVU). The NVU comprises endothelial cells, pericytes, astrocytes, microglia, neurons, and smooth muscle cells, working together to regulate cerebral homeostasis [162, 163] (Fig. 3).

Within the NVU, highly specialised endothelial cells form the physical interface between neural tissue and the bloodstream. Brain endothelial cells establish tight junctions at the interendothelial contacts, creating a selective barrier that limits paracellular diffusion. These junctions consist of protein complexes including claudins, occludins, tricellulins, junctional adhesion molecules (JAMs), and zonula occludens (ZO) proteins. The overall permeability of the BBB is regulated not only by the integrity of these junctions but also by NVU components such as pericytes, astrocyte end-feet, and microglia.

Astrocytes are known to modulate BBB permeability in mammals [164], while microglia activation and subsequent inflammatory response contribute to BBB dysfunction [165]. Pericytes envelop of the endothelial surface varies from 40 to 80% depending on the species and region analysed [166–168], whereas astrocytic end-feet coverage estimates range up to 100%, forming a highly interdigitated and continuous interface with the vasculature [169]. These end-feet are enriched with both small, round, and elongated mitochondria, reflecting high metabolic activity and supporting their role as a dynamic metabolic barrier at the neurovascular interface [166]. Astrocytes also contribute to neurovascular coupling by modulating vascular tone in response to neuronal metabolic demands, a process largely driven by glutamate release. This regulation involves the release of potassium ions from astrocytic end-feet into the perivascular space, leading to arteriolar vasodilation [170].

Importantly, astrocytes are a heterogeneous cell population, and their identification requires more than just canonical markers such as GFAP. They vary in lineage, proliferation capacity, ion channel expression, gap junction connectivity, electrophysiological properties [171], and calcium signalling dynamics. In the grey matter, protoplasmic astrocytes occupy discrete, non-overlapping domains [172]. For instance, in the hippocampus, a single astrocyte can contact up to 140,000 synapses [172] and likely regulates the vasculature tone within its territorial domain, highlighting its integrative role in both synaptic and neurovascular regulation.

Beyond the NVU, additional brain barriers play crucial roles in regulating molecular exchange. The blood-cerebrospinal fluid barrier (BCSFB) is located at the choroid plexus in the lateral, third, and fourth ventricles, while the arachnoid barrier in the meninges further isolates the brain from the systemic circulation [13, 164]. At the choroid plexus, a single layer of villous epithelial cells encapsulates immune cells, stromal cells, and an extensive fenestrated capillary network, forming a highly selective diffusion barrier while also secreting

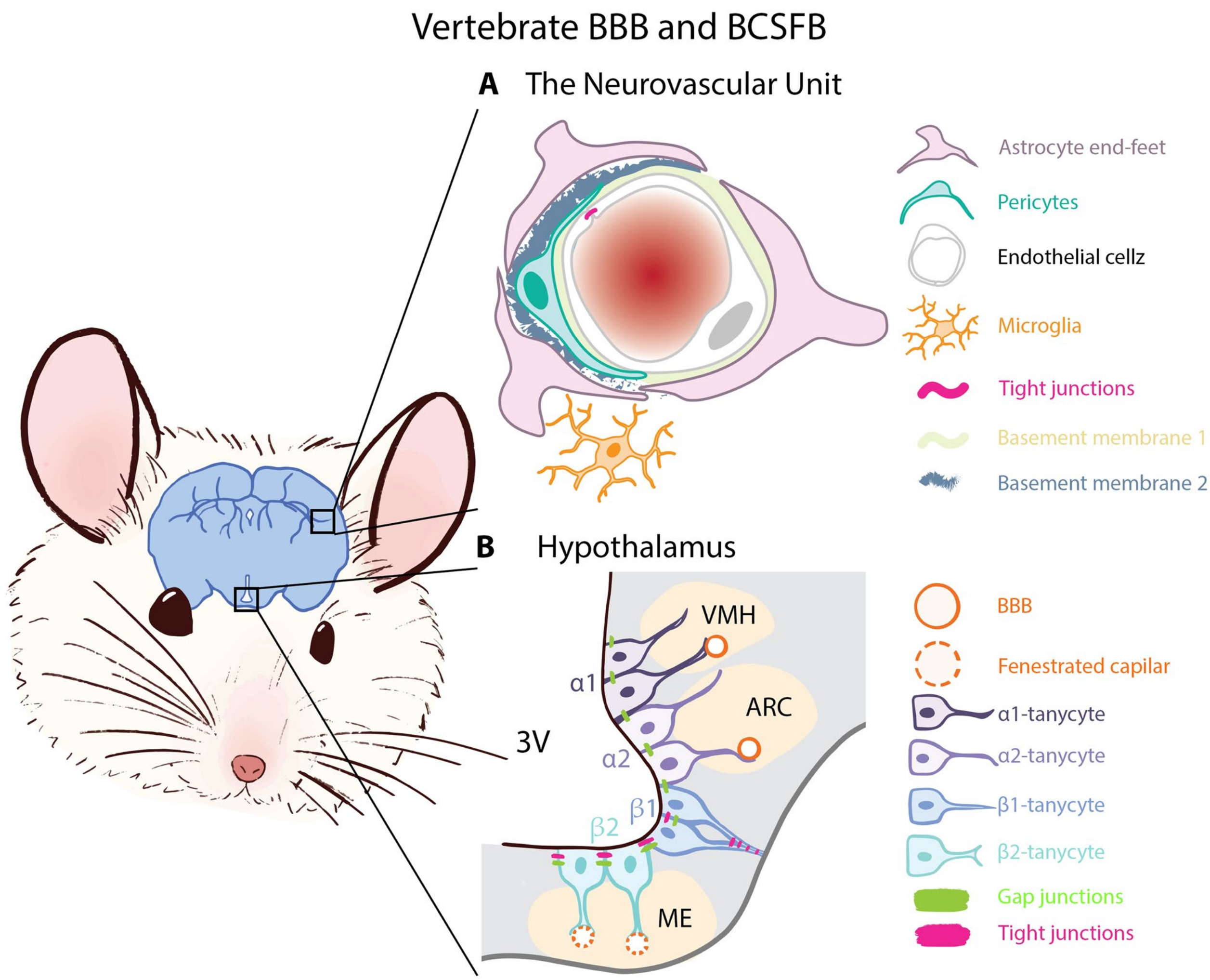

Fig. 3 The vertebrate blood–brain and blood-cerebrospinal fluid barriers. A representative coronal section of the mouse brain. **A** Diagram of the neurovascular unit (NVU), which consists of microglia, astrocyte end-feet, pericytes, and endothelial cells. Tight junctions are established between endothelial cells of the brain blood vessels, maintaining the structure and function of the blood–brain barrier (BBB). **B** Schematic representation of hypothalamic tanycytes at the floor of the third ventricle (3 V), where β2-tanycytes form the blood-cerebrospinal fluid barrier (BCSFB). Additionally, β1-tanycytes establish a parenchymal barrier that separates the median eminence (ME) from the arcuate (ARC) and ventro-medial hypothalamic nuclei (VMH). α-tanycytes generate a non-occluding CSF-parenchymal barrier

and mobilising cerebrospinal fluid (CSF) [173, 174]. Similarly, arachnoid cells create a tight junction-sealed interface that isolates the CSF-filled subarachnoid space from the fenestrated blood vessels of the dura mater [164, 175].

In addition to these barriers, the vasculature of the circumventricular organs, often referred to as the "*windows of the brain*", is fenestrated and lacks typical BBB functions. An exemplary case of glial cells contributing to barrier formation in these regions is the tanycytes—a specialised subset of ependymal cells of glial origin—that participate in the blood–cerebrospinal BCSFB at the median eminence (ME), regulating the influx and efflux of macromolecules.

Tanycytes maintain the CSF-blood barrier homeostasis

Tanycytes (from the Greek *tanus*: elongated) are specialised ependymal cells arranged in a monolayer lining the lateral walls and floor of the third ventricle (3V), near hypothalamic nuclei. Unlike common ependymal cells, tanycytes have only one or two cilia on their apical surface, which slows down the CSF flow in this cavity [176]. Their bipolar morphology resembles that of embryonic radial glia cells, with proximal poles contacting the CSF and single distal extensions projecting toward the arcuate (ARC) and ventromedial nuclei, some reaching the pial surface [177].

The lineage of tanycytes has been a subject of debate, as they exhibit characteristics of both astrocytes and embryonic radial glia. Some subpopulations express typical astrocytic markers, such as GFAP and GLAST [178], but also embryonic radial glia typical markers, such as Rax, Lhx2, and Sox2 transcription factors, the intermediate filament proteins Vimentin and Nestin, and the brain lipid binding protein (BLBP). Moreover, tanycytes also retain the ability to proliferate and differentiate into mature glia and neurons in an adult organism [179–182]. Notably, the embryonic origin of tanycytes can be traced as early as embryonic day 11 (E11) in mice, occurring in parallel with neurogenesis [183]. This process involves a direct transition of a subpopulation of radial glia cells, which exit the cell cycle, into tanycytes [183].

While tanycytes are found in birds and fish (with variations in marker expression and proliferative ability) [184], most of their characterisation has been focused on mammals. Thus, tanycytes are broadly categorized into α-tanycytes, which Line the lateral walls of the 3V, and β-tanycytes, whose cell bodies reside in the floor of the 3V and extend projections that branch into thin protrusions terminating at the fenestrated capillaries of the ME, a circumventricular organ [185]. Additionally, fibroblast growth factor receptor 1 (FGFR1) regulates the length of β-tanycyte processes, as its genetic ablation results in shorter projections [186].

The expression of ZO-1 and claudin-1 forms organised tight junctions between β-tanycytes cell bodies, shaping the BCSFB [176, 187, 188]. The extension of these tight junctions is highly dynamic, and modifiable in response to the metabolic status of the animal [189]. Thus, α-tanycytes form a non-occluding barrier between the cerebrospinal fluid (CSF) and the brain parenchyma, while β2-tanycyte barrier restricts the free diffusion of molecules from the fenestrated capillaries of the ME into the CSF and the surrounding parenchyma (Fig. 3B).

Due to their strategic location, tanycytes act not only as a barrier but also as neuromodulatory cells, regulating hormone availability and access from peripheral blood and/or CSF to ARC neurons. For instance, tanycytes influence energy homeostasis by controlling access to leptin, which induces satiety [190], and ghrelin, which triggers hunger [191, 192]. The processes of β2-tanycytes control the release of hypothalamic neuropeptides, such as thyrotropin-releasing hormone (TRH) [193] and gonadotropin-releasing hormone (GnRH), into the portal vasculature of the ME, a process modulated by nitric oxide and hormones [194, 195]. This tanycyte–GnRH neuron–portal capillary complex is referred to as the "GnRH-tanycyte unit" [196].

Tanycytes also modulate animal metabolism and food intake. They release FGF21 as a paracrine neuronal signal in response to peripheral palmitate, inducing satiety and regulating lipid homeostasis [197]. Moreover, tanycytes express the GLUT2 transporter, which has a high capacity for glucose, allowing them to detect and respond to fluctuations in blood sugar levels [198]. Similar to the *Drosophila* subperineurial glia, tanycytes express monocarboxylate transporters (MCT1 and MCT4) [199], which mediate lactate release, and directly influence the activity of neurons controlling energy balance [200]. The metabolic relevance of tanycytes has been demonstrated in genetically modified mice lacking α- and β-tanycytes, which exhibit increased visceral fat distribution and insulin resistance [201]. Additionally, tanycytes participate in thyroid hormone metabolism, converting thyroxine (T4) into the active triiodothyronine (T3) within the brain [202].

Both α- and β-tanycytes are extensively interconnected through gap junctions [203], specifically by Connexin-43 (Cx43), forming a panglial coupled network that extends to parenchymal astrocytes and oligodendrocytes [204]. This network enables the propagation of calcium waves in response to stimuli such as high-glucose levels [205] and non-nutritive sweeteners [206]. Moreover, Cx43 expression in tanycytes is associated with their responsiveness to purinergic signals, which trigger the generation of calcium waves [207] and promote cell proliferation [208].

The strategic position of β-tanycytes at the blood-CSF interface, along with their role in regulating the transport of signalling molecules between the fenestrated capillaries of the ME, the surrounding hypothalamic parenchyma, and the CSF, underscores their critical function as a specialised barrier that integrates metabolic, neuroendocrine, and neuromodulatory processes. Interestingly, tanycytes resemble the subperineurial glial barrier in insects, including their capacity to sense the animal's nutritional status and extend membrane projections [25, 87].

The perivascular glial cells form the blood–brain barrier in select fish

In most vertebrates, the BBB is of the endothelial type. However, two subclasses, *Elasmobranchii* (sharks, skates, and rays) and *Chondrostei* (sturgeon), are the exception to this pattern [209]. Early studies described the ultrastructure of the shark BBB to be different from mammals, observing that perivascular glial cells surround brain blood vessels and frequently make direct contact with them [210]. Later, Paulo Hashimoto identified astrocyte end-feet as forming the barrier and preventing the influx of horseradish peroxidase (HRP) [211]. In these species, tight junctions in perivascular glia restrict paracellular diffusion, rather than endothelial cells [212]. A similar glial barrier has been observed for sturgeons [209], suggesting that an ancestral endothelial barrier function has

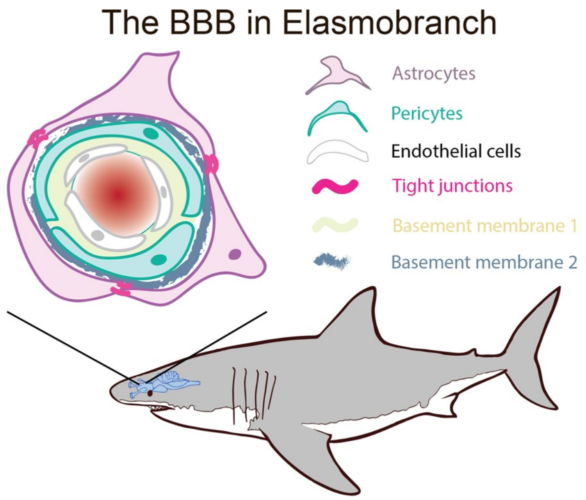

Fig. 4 The shark blood–brain barrier. In elasmobranchs and chondrostean fish, the blood–brain barrier (BBB) is established at the glial level. Brain blood vessels are fenestrated, while astrocytes end-feet establish tight junctions that effectively block paracellular diffusion of macromolecules. See of Bundgaard and Cserr [212] Figs. 4 and 5 for electron micrograph showing the ultrastructure of glial tight junctions in skates (*Raja batis, Raja erinacea* and *Raja ocellata*) [212]

been executed by the surrounding perivascular glia [22, 209, 213] (Fig. 4).

The developmental origin of this glial barrier in elasmobranchs and chondrosteans remains unknown, as well as the extent to which perivascular glia in these species perform analogous functions to endothelial cells in the vertebrate BBB. Considering structural aspects of these fish BBB, it can be hypothesised that it resembles more cephalopod glial barrier than the vertebrate endothelial barrier (Fig. 2).

Interestingly, in sharks, perivascular glia—rather than endothelial cells—express the glucose transporter GLUT1, in contrast to mammals. Additionally, monocarboxylate transporters MCT1 and MCT4 are also expressed in these glial cells [214]. This suggests that perivascular glia metabolise glucose into ketone bodies or lactate to support neuronal activity, a process analogous to the astrocyte-neuron lactate shuttle (ANLS) described in mammals [215–217]. Further support for this metabolic coupling comes from *Drosophila* [101, 113, 218–220], where both subperineurial and perineurial glia express carbohydrate transporters for glucose and trehalose, as well as monocarboxylate transporters [101, 102, 112, 221, 222]. These similarities suggest that the glial barriers of sharks and *Drosophila* may share functional parallels in the regulation of the CNS carbohydrate metabolism.

The specific type of macroglial cells structuring the BBB in sharks remains unclear. Perivascular glia have been primarily classified as astrocytic end-feet [211], but Balmaceda-Aguilera et al. suggested that these glial cells are radial glia end-feet based on their morphology and molecular markers [214]. Future studies, particularly single-cell RNA sequencing, will be essential for identifying the molecular signatures of perivascular glia and determining their similarities to invertebrate and mammalian glial cell types.

Glial cells are part of the vertebrate blood-nerve barrier

Unlike the central nervous system (CNS), the peripheral nervous system (PNS) features a distinct interface between neural tissue and the circulatory system known as blood-nerve barrier (BNB). Peripheral nerves are composed of a highly vascularised epineurium, a connective tissue sheath, rich in collagen fibrils, and an endoneurium that contains axon fascicles ensheathed by the perineurium—a structure distinct from the arthropod perineurium—and endoneurial blood vessels [52, 223, 224].

The BNB consists of two primary protective structures: endothelial cells in the endoneurial capillaries and the perineurium, which consists of a multilayered sheath of perineurial cells that encases nerve fascicles. The endothelial cells in endoneurial blood vessels form tight junctions that regulate permeability, functioning as part of the PNS neurovascular unit alongside pericytes and tactocytes, which may play a role analogous to astrocytes in the CNS [225]. Additionally, perineurial cells are also interconnected by tight junctions, acting as a barrier against toxins and immune cells while facilitating the transport of essential nutrients, ions, and metabolic substrates [224] (Fig. 5A).

During development, perineurial glial cells are derived from the neural tube floor plate, they migrate and then bundle axons into fascicles, establishing the perineurium of the BNB [224, 227, 228]. Interestingly, the vertebrate perineurium shares remarkable similarities with the *Drosophila* BNB. In flies, peripheral nerves are enveloped by three types of glial cells: wrapping glia, which encase axons, and subperineurial and perineurial glia, which constitute the barrier [66, 229–231]. From a metabolic standpoint, perineurial glia in the vertebrate BNB express the glucose transporter GLUT1, resembling the metabolic barrier role of brain and endoneurial endothelial cells [225], as well as *Drosophila* BBB glia that express the glucose transporters MFS3 and Pippin, but not Glut1 [102, 222]. These structural and metabolic resemblances suggest that the vertebrate perineurial glia and insect subperineurial glia may perform comparable barrier functions in the PNS.

Beyond its role in development, in which the vertebrate perineurium is required for Schwann cell differentiation [232], perineurial cells are essential for

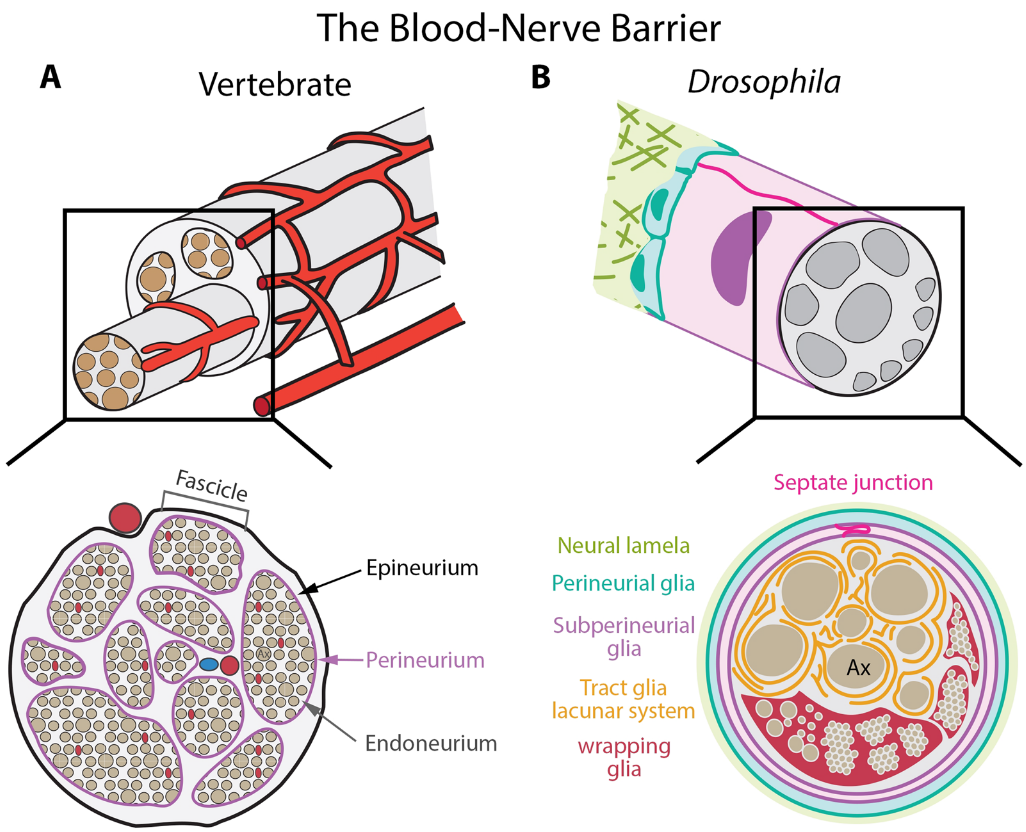

Fig. 5 The blood-nerve barrier (BNB) in vertebrates and *Drosophila melanogaster*. **A** Structure of a vertebrate nerve supplied by blood vessels. Nerve fascicles are enclosed by a glial structure known as the perineurium, where tight junctions prevent the diffusion of macromolecules. Endoneurial endothelial cells are sealed by tight junctions, forming part of the BNB. **B** In *Drosophila,* the blood-nerve barrier is established similarly to the central nervous system, with subperineurial glial cells forming septate junctions. Interestingly, each subperineurial cell forms septate junctions with itself. Tract glia create the lacunar system, a network of membranes surrounding large-diameter axons in the proximal segment of the nerve. Inside the nerve, a different type of glial cells, wrapping glia, enclose multiple axons, either individually or in groups. This wrapping resembles myelinated axons in the vertebrate peripheral nervous system [226]

nerve regeneration. Following nerve injury, distal axons undergo Wallerian degeneration [233], and perineurial cells contribute to the clearance of cellular debris [234–236]. Subsequently, perineurial glia begin bridging the proximal and distal stumps of the injured nerve, filling the injury gap in a TGF-β signalling-dependent process [236]. While *Drosophila* PNS axons have demonstrated a regenerative capacity in injury models [237], the role of subperineurial glia in this process remains unexplored. Understanding the functional parallels between vertebrate and invertebrate blood-nerve barriers will provide valuable insights into their roles in both neural homeostasis and regeneration.

Other glial barriers can isolate the neural tissue in metazoans

Glial cells exhibit an extraordinary capacity for establishing compartmental boundaries. This barrier property is not limited to isolating the circulatory system from the brain parenchyma or cerebrospinal fluid (CSF). In many cases, glial cells actively contribute to the parenchymal compartmentalisation of the nervous system, forming structural and functional boundaries that help regulate neural microenvironments.

A common example of the versatility of glial cells is their role in axonal ensheathment in vertebrate myelinated fibres. Oligodendrocytes in the CNS and Schwann cells in the PNS isolate axons, allowing saltatory conduction, a faster and more efficient way to convey information. Myelin is a multilayered membrane structure that insulates the neuronal axon in segments known as internodes, leaving short unmyelinated axonal segments called *Nodes of Ranvier* [238]. At the paranodal region, myelin lamellae establish septate-like junctions with the axon plasma membrane, anchoring the myelin layers and forming a diffusion barrier [239–241].

In the *Drosophila* CNS neuropil, there is no single-axon isolation in contrast to the peripheral nerves [229, 242]. Ensheathing glia cover the entire neuropils as whole compartments, effectively separating them from the cortex, where cell bodies reside [67, 71, 243–245]. Hence, ensheathing glia form a neuropil-cortex barrier inside the *Drosophila* CNS. Altering ensheathing glia polarisation or eliminating them causes defects in larval locomotion [246]. Although ensheathing glia do not invade the neuropil or directly regulate synaptic transmission, they play an essential role in neurotransmitter homeostasis, indirectly influencing synaptic activity [247, 248]. Thus, it is well established that isolation of inner compartments is essential for the correct function of the insect CNS.

While ensheathing glia contribute to the compartmentalization within the CNS, they do so without forming septate junctions. Instead, they function as a non-occluding/permeable barrier with extensive membrane overlap, slowing the diffusion of solutes between the cortex and neuropil, but without completely blocking it [246]. Previous studies have suggested the presence of scalariform-like junctions in glial cells at the antennal lobe [249–251]. However, the role of these adhesions in ensheathing glial barrier is unknown. Interestingly, following the ablation of ensheathing glia, another glial type (astrocyte-like glia) can partially compensate by surrounding the neuropil and replacing the missing ensheathing glial barrier [246].

As with ensheathing glia, a similar barrier function exists in the vertebrate hypothalamus between the arcuate nucleus (ARC) and the median eminence (ME). Direct injection of HRP into the third ventricle (3V) allows diffusion to the ARC but not to the ME [252]. Conversely, peripheral administration of HRP reaches the ME without affecting the ARC, suggesting the presence of a lateral physical barrier between these compartments. This parenchymal barrier is likely formed by additional tight junctions between β1-tanycyte projections [185, 253] (Fig. 3B).

Metabolite and nutrient transport across ensheathing glia has not been described yet. Notably, thin membrane processes extend from the ensheathing glial barrier towards the CNS surface, encasing neuronal cell bodies in the dorsal ventral cord and contacting subperineurial glia [87, 246]. Similarly, processes extending from subperineurial glia into the ensheathing glial barrier have also been observed [87]. These remarks support the idea that glial barriers in *Drosophila* can establish contact and communicate with one another, potentially coordinating their functions within the CNS. This hypothesis aligns with the role of astrocytes in the mammalian CNS, where astrocyte end-feet are in close contact with the endothelial BBB while also modulating neuronal communication at the synaptic level. Therefore, it is plausible that a direct interaction between BBB-associated glia and neuropil-associated glia represents a conserved feature across species.

On the other hand, *Drosophila* peripheral axons are covered by wrapping glia, which provide structural support and, in the adult, can form myelin-like sheaths, akin to vertebrate myelinating Schwann cells [226, 242, 254–256]. Insect axons are surrounded by a glial mesh-like structure referred to as the lacunar system [226, 257]. This structure is mostly formed by a subtype of glial cells, known as tract glia, at the interface between the CNS and the PNS [226, 244, 256]. Across the peripheral nerves, the lacunar systems appear to collapse and cover axons in myelin-like organisations [226, 254]. Although the role of the lacunar system is unknown, it may serve as a nutrient or ion reservoir for action potential generation at the axon initial segment, while also providing isolation from other axons in the nerve to prevent ephaptic coupling [254] (Fig. 5B).

When *Drosophila* wrapping glia are lost, axon diameter and the speed of action potential propagation are reduced, whereas the expression of the voltage-gated sodium channel, *paralytic* (*para*), is increased [226, 255]. In a recent study, wrapping glia ablation was shown to increase the diameter of both motor and sensory axons, and induce length-dependent axonal degeneration of sensory axons [258]. These findings highlight the critical role of ensheathing/wrapping glia, and also the lacunar systems, in the maintenance of axonal structural integrity and for ensuring efficient neuronal communication, suggesting that glial-mediated insulation may have evolved through convergent mechanisms in both vertebrates and invertebrates.

The evolutionary origin of glial barriers: more questions than answers

Biological barriers are an essential cellular structure in metazoan animals, enabling compartmentalisation and selective permeability across tissues. Occluding junction proteins evolved to establish tight barriers long before the emergence of complex nervous systems [41]. The most ancient phylum showing septate junction structures is Porifera (sponges) [259]. The "*barrier toolkit*" necessary for forming biological barriers can be utilised by different cellular types and organs, independently of their developmental origin, as seen in the intestinal epithelium, endothelial cells, and the epidermis. This raises an intriguing question: How did glial cells acquire the ability to form biological barriers?

The simplest explanation proposes a single origin of the BBB. Thus, the glial barrier originated before the divergence of invertebrates and vertebrates but was subsequently lost multiple times during evolution. This loss may have occurred either because a tight barrier was no longer necessary in certain invertebrate taxa or because its function was transferred to endothelial cells with the emergence of the vascular system. This evolutionary model assumes that the vertebrate endothelial barrier arose multiple times throughout evolution [23, 209, 260], something that can be considered unlikely. A more plausible scenario suggests that the BBB has independently evolved across different evolutionary lineages [22].

Given the remarkable versatility of glial cells in supporting diverse neural functions and separating compartments, it is plausible that the glial barrier evolved multiple times, adapting to the specific physiological demands of the lineage nervous system. In this context, the arthropod lineage may have developed the perineurium to regulate ion homeostasis in the nervous system, while vertebrates have repurposed glial cells multiple times to fulfil barrier functions, as seen in tanycytes and the shark BBB. Since multiple biological barriers have independently evolved to separate compartments from the circulatory system, the most straightforward explanation for the recurring emergence of barriers is the conservation of occluding junction proteins.

This barrier toolkit concept raises the question of what environmental pressures drive the selection of endothelial cells, rather than glial cells, to establish occluding cell–cell junctions. One hypothesis is based on the efficiency of transport across the barrier. In a vascularised system, positioning the barrier in direct contact with the bloodstream —via endothelial cells— may, in principle, be more effective than embedding it within astrocytes or pericytes. This arrangement allows for immediate control over molecular exchange between the blood and nervous system, optimising nutrient delivery and waste removal while maintaining homeostasis. Furthermore, the endothelial barrier permits pericytes and astrocytes to specialise in other functions, releasing them from the barrier function burden [260]. However, this idea contrasts with observations in cephalopods and elasmobranchs, where the barrier is not the first-contact tissue with the circulatory system. These differences raise further questions about the evolutionary pressures shaping the BBB organisation and how structural variations impact neural homeostasis across species.

The evolution of the glial barrier may represent a case of convergent evolution, where different glial cell types—across arthropods, cephalopods, and vertebrates—have independently adopted a shared set of molecular mechanisms to regulate the nervous system's internal environment [22]. However, this idea can be challenged by the very definition of the BBB itself. How tight must a barrier be to qualify as a BBB? Glial-ensheathed ganglia are found in simpler invertebrates, including *Helix pomatia* (*Gastropoda*) and *Lumbricus sp* (*Annelida*) [31, 261, 262]. Meanwhile, the *Caenorhabditis elegans* (*Nematoda*) exhibits sheath glial cells that ensheathe neurites of sensory neurons [263, 264]. It is possible that an ancestral glial barrier evolved early, but with varying degrees of tightness arising independently across lineages. Further comparative research across diverse phyla will be crucial for understanding how glial cells gained their barrier-forming properties and how these structures contributed to the evolution of complex nervous systems.

Conclusions

The concept of brain barriers, first introduced over a century ago, has been extensively studied across various animal groups. To date, brain interfaces have been identified in three major taxa: *Arthropoda*, *Cephalopoda*, and *Vertebrata*. These findings are largely based on the exclusion of high-molecular-weight tracers, such as horseradish peroxidase and lanthanum, from entering the CNS, or the presence of occluding junctions observed via electron microscopy. However, other invertebrate taxa may also possess leaky brain barriers that do not rely on true occluding junctions [43]. Should these barriers be considered a form of BBBs? Given the multiple functions of the human BBB, restricting the definition solely to the tightness of the paracellular barrier is overly simplistic. The BBB also regulates ion homeostasis, protects against neurotoxins, and facilitates nutrient transport. Thus, a broader definition might be necessary to encompass the diversity of brain barriers across taxa.

A deeper understanding of the function and developmental origin of invertebrate brain barriers through comparative studies remains necessary. Much of our current knowledge about glial barriers comes from model organisms like *Drosophila melanogaster*, while relatively few cellular and molecular studies have been conducted on cephalopods, chelicerates, or elasmobranchs. This knowledge gap is likely to narrow with the increasing availability of genomic data from diverse species and advancements in single-cell and single-nucleus sequencing. Identifying the transcriptomic signatures of glial cells forming brain barriers across different species will provide valuable insights into their evolutionary origins and shared features.

Abbreviations

3V	Third ventricle
ABC	ATP-binding cassette
ANLS	Astrocyte-neuron lactate shuttle
ARC	Arcuate nucleus
BBB	Blood–brain barrier
BCFB	Blood-cerebrospinal fluid barrier
BLBP	Brain lipid binding protein
BNB	Blood-nerve barrier
cAMP	3′,5′-Cyclic AMP
CNS	Central nervous system
CSF	Cerebrospinal fluid

Cx43	Connexin-43
dIlp	*Drosophila* Insulin-like peptide
EcI	Ecdysone Importer
FGFR	Fibroblast growth factor receptor
Gbb	Glass bottom boat
GFAP	Glial Fibrillary Acidic Protein
GnRH	Gonadotropin-releasing hormone
GPCR	G-protein coupled receptor
HRP	Horseradish peroxidase
IMD	Immune deficiency
JAM	Junctional adhesion molecules
MCT	Monocarboxylate transporters
Mdr	Multi drug resistance
ME	Median eminence
NVU	Neurovascular unit
scaf	Scarface
SPG	Subperineurial glia
PG	Perineurial glia
PKA	Protein kinase A
PNS	Peripheral nervous system
T3	Triiodothyronine
T4	Thyroxine
TRH	Thyrotropin-releasing hormone
ZO	Zonula occludens

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Data availability
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Declarations

Ethics approval and consent to participate
Not applicable.

Competing interests
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